

Rounds Protocol

Complete Architecture Diagrams

CHAIN

Solana

FRAMEWORK

Anchor 1.0.2

TOKEN

USDC only

INSTRUCTIONS

13

ACCOUNT TYPES

8

MAX MEMBERS

20 per circle

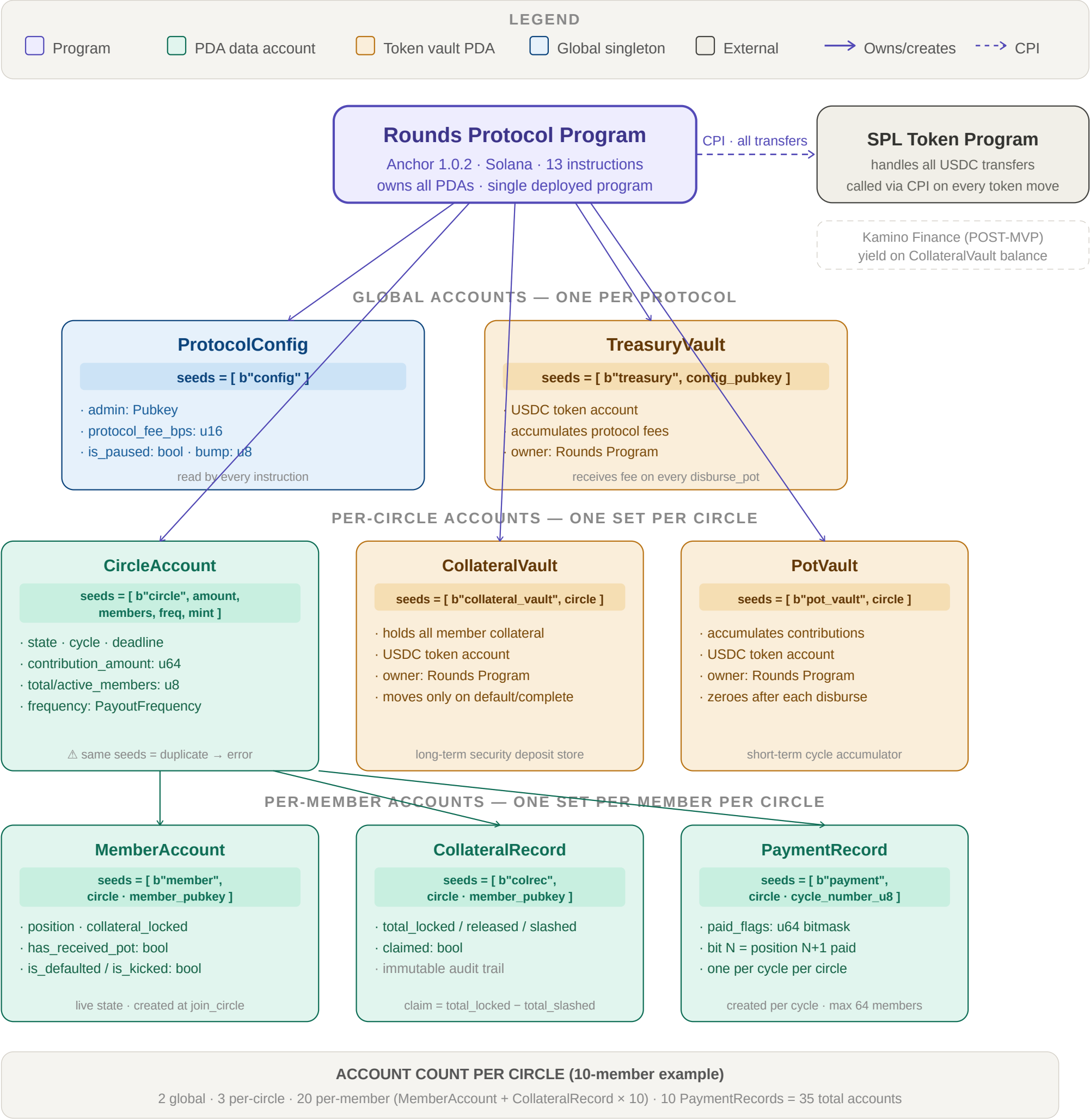
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Rounds Protocol is a decentralised trustless ROSCA on Solana.
A sliding-scale collateral mechanism mathematically eliminates default risk.

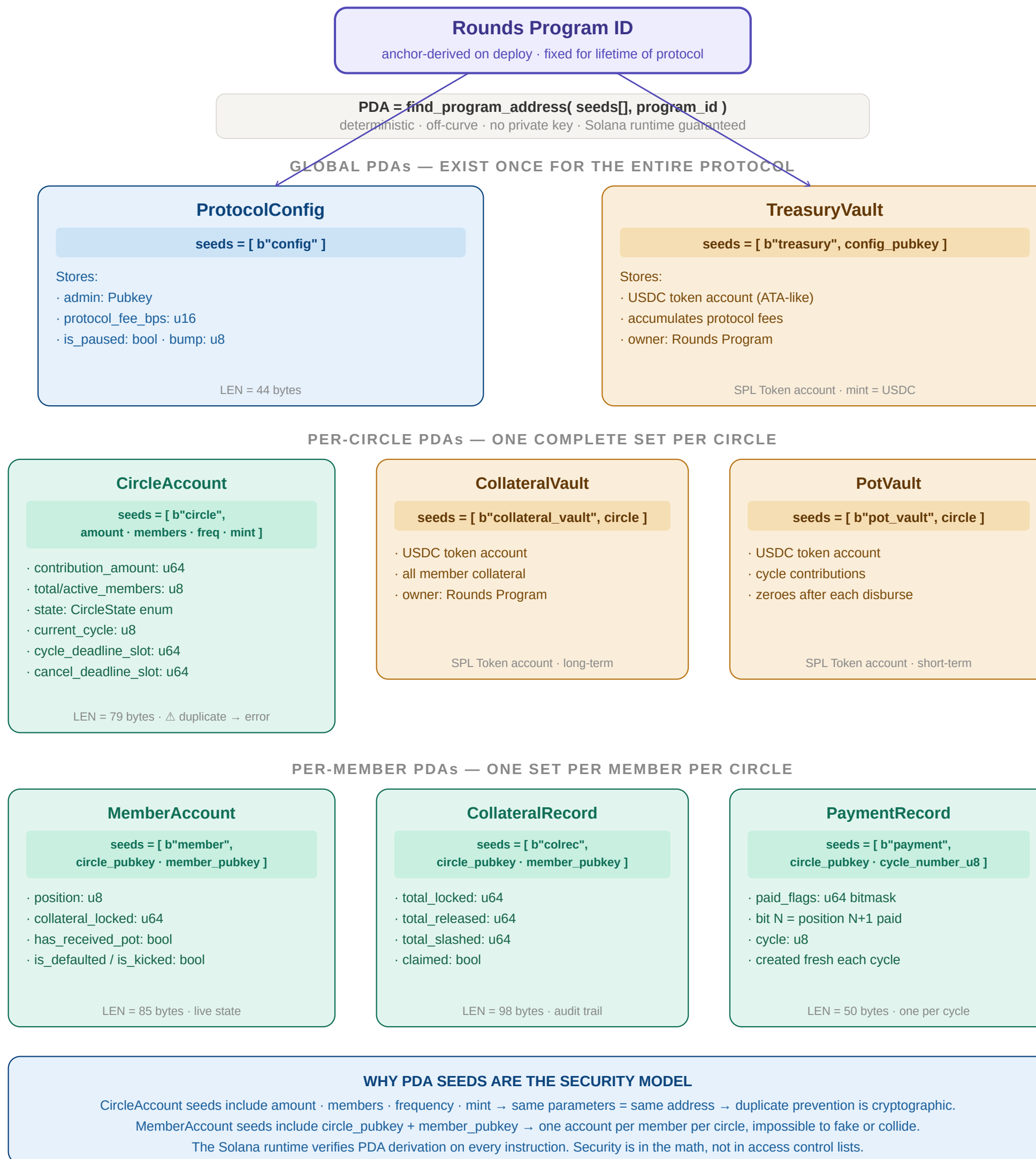
Program & Account Overview

The Rounds Program as the central entity — all account types, their ownership, and the SPL Token CPI relationship.



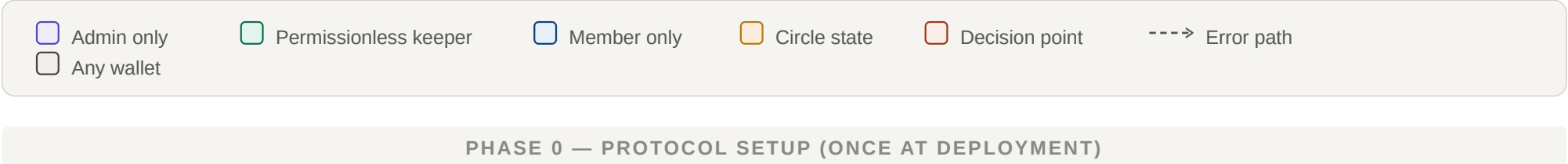
PDA Derivation Map

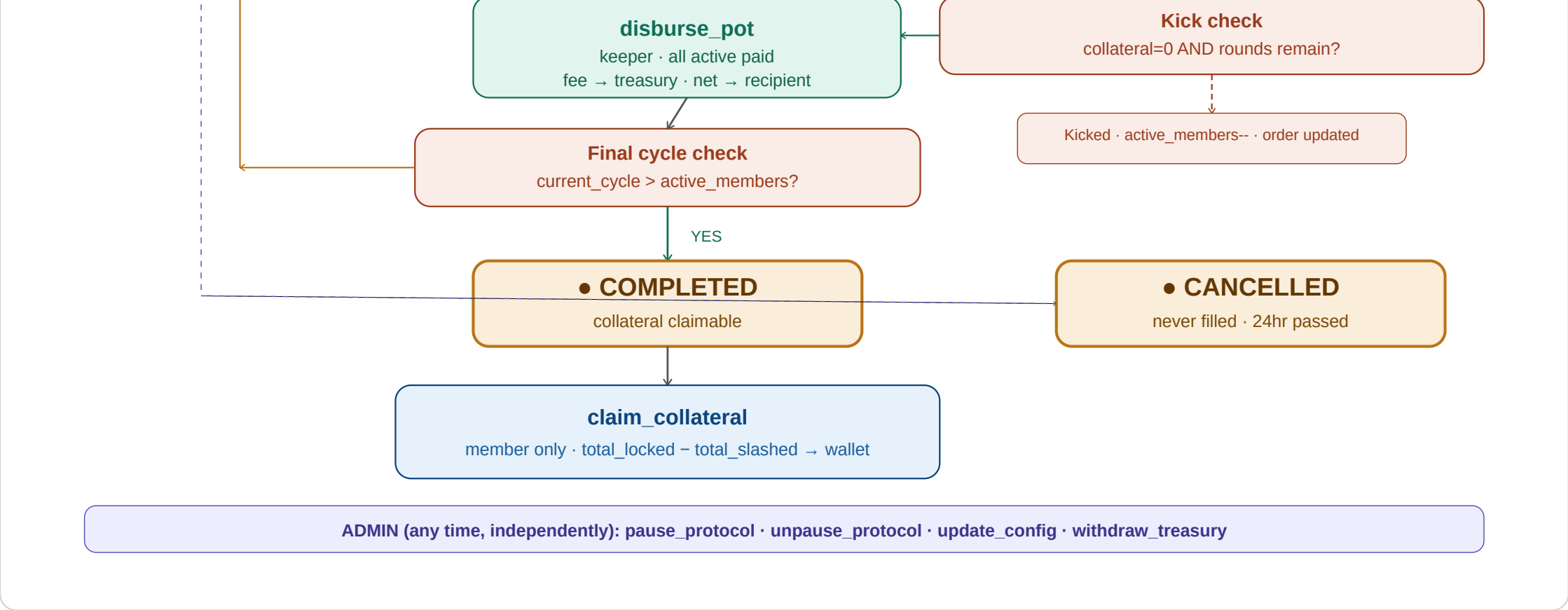
Every PDA with its exact seeds, stored data, and security model. Seeds + Program ID → unique deterministic address. No private key controls any PDA.



Instruction Flow & State Machine

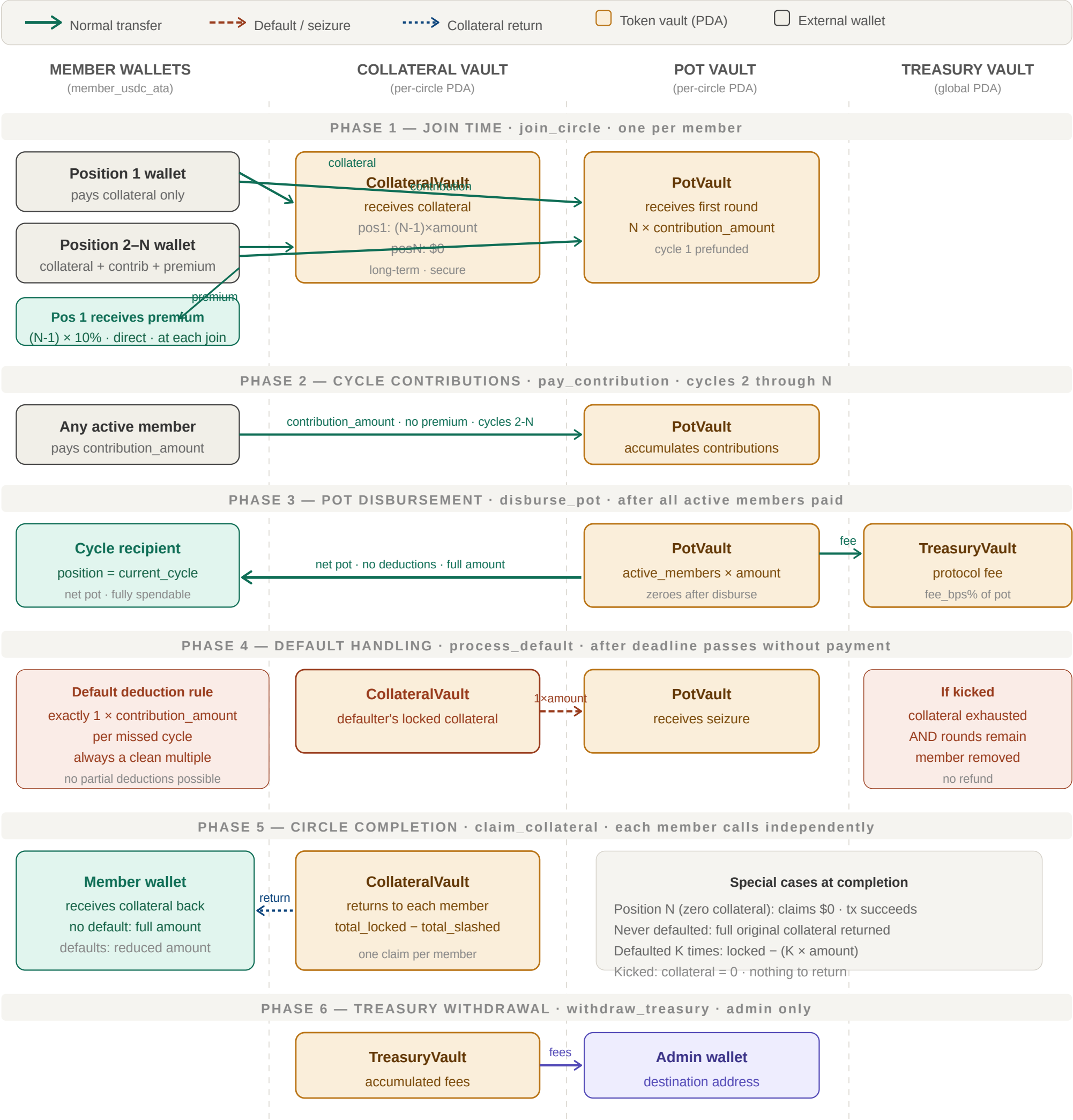
All 13 instructions connected through the circle lifecycle. Every state transition, decision point, error path, and permission level shown.





Token Flow Diagram

Every USDC movement across the full circle lifecycle. All transfers executed via CPI to SPL Token Program.



AT JOIN: Each member sends collateral → CollateralVault · \$100 → PotVault · 10% premium → Position 1 wallet (positions 2-N)

EACH CYCLE: Each member sends \$100 → PotVault (no premium from cycle 2 onwards)
EACH DISBURSE: PotVault → fee to TreasuryVault · net pot → cycle recipient wallet · PotVault zeroes completely
ON DEFAULT: CollateralVault → exactly \$100 → PotVault · circle continues · kicked if collateral = 0 AND rounds remain
AT COMPLETION: CollateralVault → (total_locked – total_slashed) → each member wallet · TreasuryVault → admin on withdraw

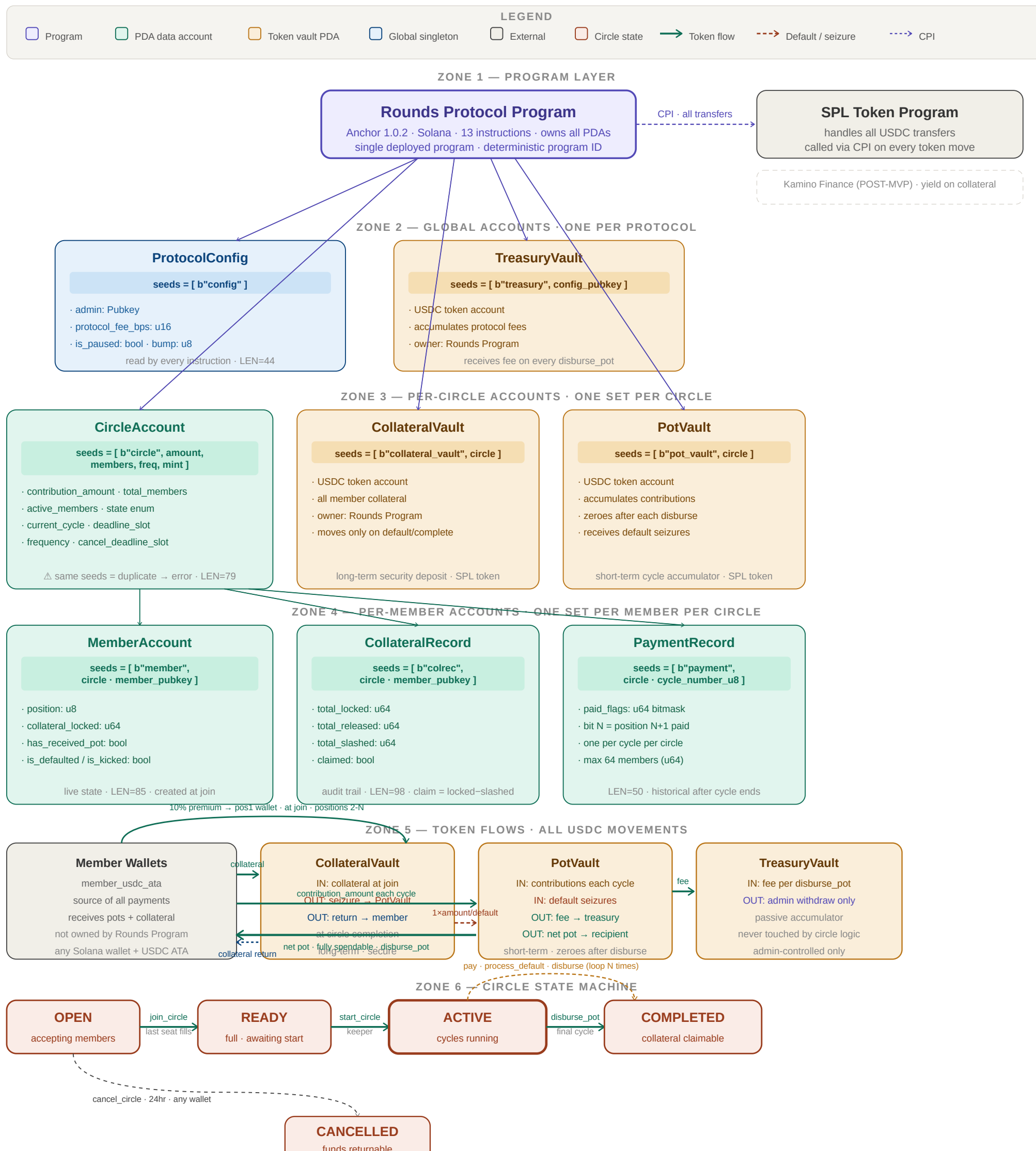
CORE TOKEN INVARIANT — ENFORCED AT ALL TIMES

$\text{collateral_locked} \geq \text{remaining_cycles} \times \text{contribution_amount}$ · checked on every state transition

Collateral is always a clean multiple of contribution_amount · no partial deductions possible · defaulting is always irrational

Consolidated Architecture

The complete protocol in one reference diagram — program structure, all accounts with PDA seeds, state machine, token flows, and instruction directory.



☐ ADMIN ONLY

initialize_protocol — deploys ProtocolConfig + TreasuryVault · once at launch
 pause_protocol — sets is_paused = true · halts all instructions immediately · funds stay in vaults
 unpause_protocol — sets is_paused = false · resumes all operations immediately
 update_config — updates fee_bps · applies to future disbursements only
 withdraw_treasury — moves accumulated fees from TreasuryVault to admin wallet

☐ PERMISSIONLESS KEEPER — any wallet · same pattern as Aave/Kamino liquidation bots

create_circle — initialises CircleAccount PDA · duplicate open circle → descriptive error · state = OPEN
 start_circle — READY → ACTIVE · sets cycle 1 deadline · inits PaymentRecord · cycle 1 prefunded at join
 disburse_pot — all active paid → fee to treasury · net pot to recipient · cycle++ · ACTIVE → COMPLETED on final
 process_default — deadline passed + unpaid → deduct 1×amount from collateral · kick if collateral=0 + rounds remain
 cancel_circle — OPEN + 24hr timeout → CANCELLED · collateral and contributions returnable to all joined members

☐ MEMBER ONLY — requires valid MemberAccount for that circle

join_circle — assigns position by join order · collateral → CollateralVault · contribution → PotVault · premium → pos1
 pay_contribution — cycles 2-N · exact contribution_amount → PotVault · updates PaymentRecord bitmask
 claim_collateral — COMPLETED · total_locked – total_slashed → member wallet · one independent call per member

CORE INVARIANT: collateral_locked ≥ remaining_cycles × contribution_amount · enforced on every state transition
 Collateral is always a clean multiple of contribution_amount · no partial deductions possible · defaulting is always irrational